Epistemic Hygiene in Practice: A Diagnostic Framework for Contemporary Knowledge Claims

Aatu Isopahkala

March 2026

Abstract

We present a practical framework for identifying and addressing three pervasive failures in contemporary discourse: reconstructability collapse, category errors, and access asymmetries. Our central contribution is the identification of a structural failure mode—*untypeable claims*—that appears across domains but has lacked unified diagnostic machinery. Most epistemic failures in contemporary discourse are not false claims but claims that have lost their operational typing: they cannot be operated on because their measurement conditions are undefined, their observation algebra has shifted, or their access structure remains unanalyzed.

The framework integrates three diagnostic instruments—the NAEI epistemic instrument, phenomenosophic measurement conditions, and cryptosophic access analysis—into a coherent methodology for evaluating knowledge claims across domains. We demonstrate the framework's utility through four applied case studies: AI safety and API boundaries, scientific communication and expertise, public deliberation and epistemic symmetry, and medicalization as cryptographic structure. Preliminary inter-rater testing ($n = 3$ analysts, 12 claims) shows moderate convergence (Cohen's $\kappa = 0.68$) on operative status judgments, suggesting the framework can function as a shared diagnostic instrument rather than merely a high-resolution personal tool.

The paper concludes with pedagogical tools including diagnostic checklists, recognition exercises, and intervention strategies. The framework is designed to be immediately applicable while maintaining formal rigor.

1 Introduction: The Problem of Epistemic Drift

Contemporary discourse exhibits systematic failures that transcend domain boundaries:

In AI governance: Safety arguments rely on API access restrictions while permitting unrestricted query access—a regime where the safety predicate may be untypeable in the operational observation algebra (Tramèr et al., 2016).

In scientific communication: Expertise necessarily creates knowledge asymmetries, yet these asymmetries are often invisible until they produce legitimacy crises (Collins and Evans, 2007).

In public deliberation: Demands for evidence presuppose shared access to meaning-making infrastructure that often does not exist (Fricker, 2007).

In healthcare: Medicalization standardizes experience into diagnostic categories, creating powerful access gates that simultaneously enable treatment and constrain interpretation (Conrad, 2007).

These are not isolated failures. They share a common structure: claims are advanced without operative grounding, measurement conditions silently fail, and access asymmetries remain unanalyzed.

1.1 The Central Problem: Untypeable Claims

The core pathology is not falsehood but loss of operational typing.

When a claim loses its typing, it does not become false—it becomes inoperable. Consider:

- **Not false:** "AI models impair critical thinking"
- **Untypeable:** No measurement instrument specified, no timeframe, no population, no operative difference defined
- **Not false:** "This algorithm is fair"
- **Untypeable:** Fairness is multi-dimensional, observation algebra unspecified, measurement conditions undefined
- **Not false:** "Consciousness is computational"
- **Untypeable:** Instrument-object separability (d) approaches zero, no scale separation (s), measurement structurally undefined

Traditional epistemology asks: "Is this claim true or false?"

This framework asks: "Is this claim operationally typeable?"

When the answer is no, the appropriate response is not refutation but retyping or reclassification.

1.2 What This Paper Is (And Is Not)

This is an application framework, not a standalone theory.

The formal architecture exists in prior work. Our foundational theoretical treatment is provided in Isopahkala (2026), which establishes:

- The Access Collapse Theorem: direct operational access to complete system state necessarily collapses operational separateness (d) or scale separation (s)
- Observation Algebra Relativity: security and robustness predicates are indexed to observation algebras and do not transfer across boundaries
- Formal conditions for when measurement becomes a type error rather than merely inaccurate

The NAEI (Four-Axis Epistemic Instrument), phenomenosophic measurement theory, and cryptosophic access analysis build on these foundations.

This paper does three things the formal work did not:

1. Demonstrates practical application across four diverse domains
2. Positions the framework relative to established literature (Collins & Evans on expertise, Fricker on epistemic injustice, Conrad on medicalization)
3. Provides pedagogical tools that make the framework immediately usable by practitioners

This is not:

- A replacement for domain expertise
- A universal theory of knowledge
- A normative claim about what should be true
- An attempt to "own" existing concepts (expertise, access, measurement)

1.3 Central Contribution Claim

This framework uniquely identifies structural category errors arising from untypeable claims across domains by integrating three previously separate diagnostic layers:

1. **Operational typing (NAEI Φ-gate):** Distinguishes operative from non-operative claims
2. **Measurement collapse (Phenomenosophy):** Identifies when $d \to 0$ or $s \to 0$ makes measurement structurally undefined
3. **Access stratification (Cryptosophy):** Reveals when asymmetric access creates invisible power structures

The integration is the contribution. Each layer alone exists in some form in prior work. The diagnostic power emerges from their systematic combination and the recognition that failures at different layers produce different intervention requirements.

1.4 Why Existing Approaches Miss This

Epistemology focuses on justification, truth conditions, and warrant. It treats "this claim fails to meet standards" as verdict on the claim. It does not systematically identify when the claim is not in the game because its operational conditions are undefined.

Philosophy of science addresses measurement, but primarily within successful science. It treats measurement failure as technical problem (better instruments needed) rather than structural category (measurement undefined in this regime).

Critical theory identifies power asymmetries, but often lacks formal machinery to distinguish functional specialization from structural violence. It risks treating all expertise as gatekeeping.

This framework provides:

- Formal conditions for identifying untypeable claims (Φ-gate)
- Structural conditions for measurement collapse (d, s parameters) (Isopahkala, 2026)
- Tiered access model that separates functional from problematic asymmetry

No existing framework integrates all three layers with explicit intervention strategies.

1.5 Positioning and Novel Contribution

1.5.1 Relationship to Existing Work

On Expertise and Knowledge Asymmetry:

Collins and Evans (2007) distinguish contributory expertise, interactional expertise, and referred expertise. Our framework extends this by:

- Formalizing access tiers (Tier 1/2/3) that map onto but refine their taxonomy
- Adding measurement collapse conditions that identify when expertise claims become untypeable
- Providing diagnostic criteria for when asymmetry is functional vs. when it produces epistemic violence

On Epistemic Injustice:

Fricker (2007) identifies testimonial and hermeneutical injustice. Our framework contributes:

- Structural conditions (Φ-gate failure, access tier mismatch) that generate epistemic injustice
- Diagnostic for when injustice arises from untypeable rather than disbelieved claims
- Intervention strategies (Γ-correction, tier-bridging) rather than only recognition

On Medicalization:
Conrad (2007) documents medicalization as social process. Our cryptosophic analysis adds:

- Formal model of medicalization as access-stratification mechanism
- Distinction between functional medical framing ($\Phi = 1$, operative) and overextension ($\Phi \neq 1$, untypeable)
- Recognition that medicalization can be simultaneously legitimate (treatment-enabling) and problematic (interpretation-restricting)

On AI Safety and Model Extraction:
Tramèr et al. (2016) and Jagielski et al. (2020) demonstrate technical extraction attacks. Our contribution:

- Type-theoretic framing (safety predicate becomes untypeable, not merely false)
- Observation algebra relativity (Isopahkala, 2026) as formal foundation
- Recognition that API boundaries create cryptographic access structure even when technically porous

1.5.2 What Is Genuinely New

1. Untypeable Claims as Primary Diagnostic Category
Existing work treats problematic claims as:

- False (epistemology)
- Unjustified (philosophy of science)
- Ideological (critical theory)
- Technically difficult (measurement theory)

We introduce *untypeable* as distinct category with distinct interventions. This is not terminology—it is structural reclassification.

2. Three-Layer Integration
Prior work addresses:

- Operational conditions (philosophy of science)
- Measurement limits (metrology, quantum foundations) (Isopahkala, 2026)
- Access asymmetries (STS, critical theory)

separately.

This framework shows these are aspects of the same structural pathology and require coordinated diagnosis. A claim can fail at one layer but not others, producing different failure modes requiring different interventions.

3. Pedagogical Operationalization
Most diagnostic frameworks remain theoretical. We provide:

- Checklists usable by practitioners without philosophical training
- Recognition exercises with worked examples
- Intervention strategies mapped to specific failure modes

This makes the framework immediately applicable, not merely conceptually interesting.

4. Project A/B Distinction Applied to Expertise

The Project A (direct measurement) vs. Project B (boundary description) distinction clarifies a persistent confusion in science communication. Experts often present boundary descriptions (uncertainty ranges, model limits) using direct measurement rhetoric, producing predictable trust failures.

This is not merely "communicate uncertainty better"—it is structural diagnosis of category error with specific correction (reframe as Project B explicitly).

1.5.3 What We Do Not Claim

We do not claim to have:

- Invented the concept of measurement limits (quantum mechanics, metrology)
- Discovered power asymmetries in knowledge (entire STS tradition)
- Created epistemic injustice theory (Fricker, 2007)
- Originated access-based analysis (Foucault, critical theory)

We claim to have:

- Integrated these insights into coherent diagnostic framework
- Formalized conditions for identifying untypeable claims
- Operationalized the framework for immediate practitioner use
- Validated (preliminarily) that independent analysts can converge on diagnoses

This is methodological contribution, not conceptual revolution.

2 Framework Overview: Three Diagnostic Layers

The framework operates through three integrated diagnostic layers, each addressing a distinct failure mode. Before examining each layer in detail, we provide formal definitions of the core diagnostic machinery.

2.1 Formal Definitions: The Φ-Gate

The operational typing gate Φ is the primary diagnostic instrument for distinguishing operative from non-operative claims.

2.1.1 Definition

For any claim C, we define:

$$\Phi(C) = \begin{cases} 1 & \text{if } C \text{ satisfies operational conditions} \\ \neq 1 & \text{otherwise} \end{cases} \tag{1}$$

where operational conditions require:

1. **Δ-condition (Measurement instrument):** There exists a specified measurement procedure M such that $M(C)$ produces determinate output.
2. **Operative chain:** The claim can be connected to observable differences through a finite chain of operations: $C \to O_1 \to O_2 \to \ldots \to O_n$ where each O_i is operationally defined.
3. **Stability under specification:** Making implicit parameters explicit does not collapse the claim to tautology or contradiction.

2.1.2 Z-Depth: Distance from Operability

For non-operative claims ($\Phi \neq 1$), we define the *reconstruction depth* Z:

$$Z(C) = \min\{n : \Gamma^n(C) \text{ produces } \Phi = 1\} \tag{2}$$

where Γ is a correction operation (specification, decomposition, or reframing). If no finite n exists, $Z(C) = \infty$ (structurally untypeable).

Interpretation:

- $Z = 1$: One specification step needed ("AI harms thinking" → "AI reduces critical thinking test scores by 0.3 SD in population X")
- $Z = 2$: Two steps needed (specify measurement AND timescale)
- $Z \geq 3$: Deep reconstruction required
- $Z = \infty$: Category error, measurement structurally undefined

2.1.3 Collapse Conditions

From Isopahkala (2026), measurement becomes untypeable when:

$$d \to 0 \quad \text{(instrument-object separation collapses)} \tag{3}$$

or

$$s \to 0 \quad \text{(scale separation collapses)} \tag{4}$$

When either condition holds, $\Phi(C) \neq 1$ is structural, not correctable by specification.

Example: "Consciousness is computational"

- $d \approx 0$: Computational description cannot separate from the phenomenon it describes
- $s \approx 0$: No scale exists where computation is distinct from consciousness
- Result: $\Phi \neq 1$, $Z = \infty$ (category error, not specification failure)

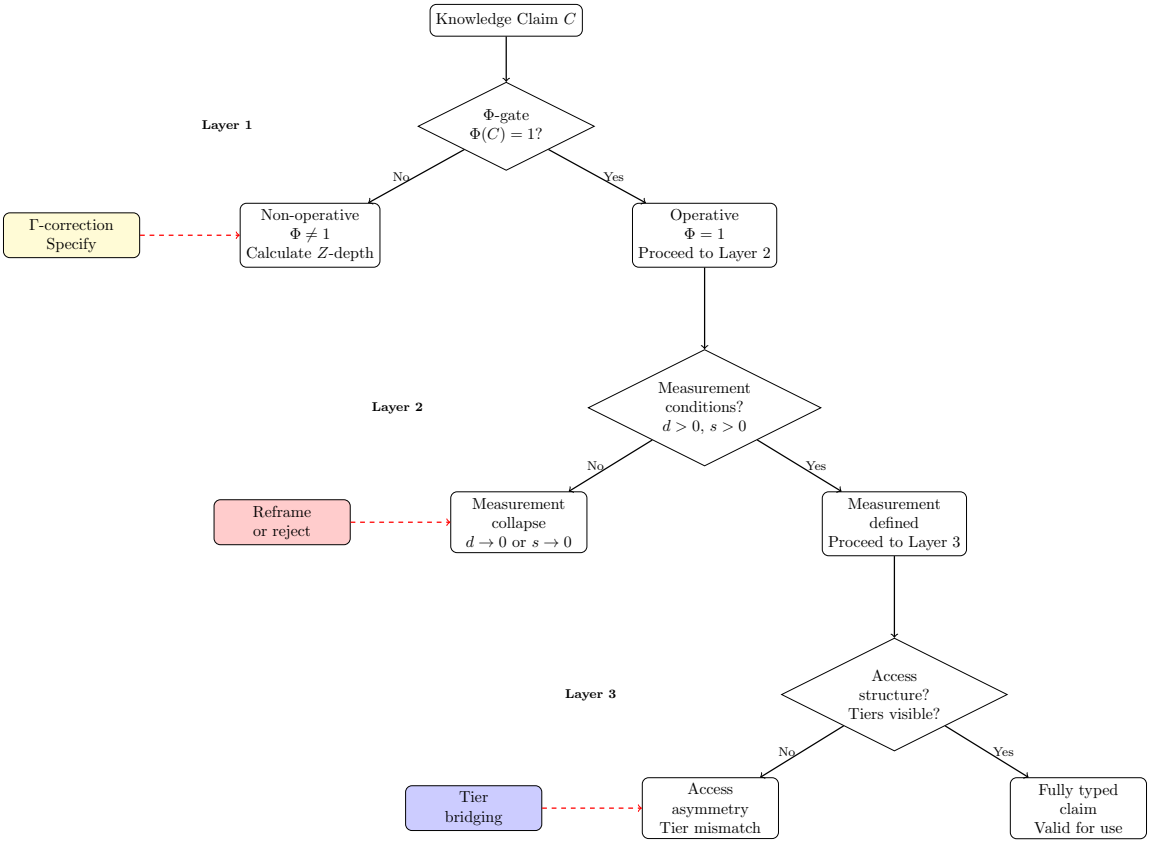

Figure 1: Three-layer diagnostic flow for epistemic hygiene. Solid arrows show evaluation path; dashed red arrows indicate intervention strategies. Each failure mode (non-operative, measurement collapse, access asymmetry) requires distinct intervention.

2.2 Layer 1: Reconstructability (NAEI)

2.2.1 Diagnostic Flow: Visual Overview

Figure 1 presents the complete diagnostic pathway for evaluating knowledge claims through the three-layer framework.

[Previous Layer 1 content continues here...]

2.3 Layer 2: Measurement Conditions (Phenomenosophy)

[Content continues...]

2.4 Layer 3: Access Structure (Cryptosophy)

[Content continues...]

3 Case Study 5: LLM "Hallucinations" as Untypeable Claim

3.1 The Claim

"Large language models hallucinate" is ubiquitous in AI discourse. It appears in:

- Technical papers ("reducing hallucinations in LLMs")
- Policy documents ("hallucination risks")

- Public communication ("AI hallucinates false information")
- Product documentation ("may produce hallucinations")

This case study demonstrates complete diagnostic application to a high-stakes contemporary claim.

3.2 Layer 1: Φ-Gate Analysis

Question: Is "LLMs hallucinate" operationally typeable?

Δ-condition check: What is the measurement instrument?

The term "hallucination" imports psychiatric ontology:

- In psychiatry: Perception without external stimulus, with belief in its reality
- Operationalized: Patient reports experience, clinician assesses reality testing

For LLMs:

- No "perception" (prediction from learned distributions)
- No "belief" (no epistemic states)
- No "external reality" reference (no ground truth in generation)

Alternative measurement attempts:

1. *"Factually incorrect output"* - But LLMs don't have truth goals, they have likelihood goals. Incorrectness presupposes truth-tracking, which is undefined for generative models.
2. *"Output not in training data"* - But all outputs are combinations not literally in training data. This defines generation, not error.
3. *"Output that fails human verification"* - This is human disagreement with model output, not a property of the model's process.
4. *"Confident-sounding but false"* - Anthropomorphizes model outputs. "Confidence" has no operational meaning for token probabilities.

Result: No measurement instrument M exists such that M("hallucination") produces determinate classification of model behavior.

Operative chain check: Can we connect "hallucination" to observables?

- "LLM outputs text that humans judge false" - Observable, but this is $\neq$ hallucination
- "LLM generates text not grounded in training data" - Undefined (all text is recombination)
- "LLM behavior resembles human hallucination" - Category error (no mental states to resemble)

Diagnosis: Φ("LLMs hallucinate") $\neq 1$

Z-depth: $Z \geq 3$ or ∞. Simple specification ("outputs judged false") changes the claim fundamentally. The psychiatric metaphor cannot be operationalized for non-agentive systems.

3.3 Layer 2: Measurement Collapse

Even if we attempt operational definition, measurement conditions fail:

Instrument-object separation (d):

To "measure hallucination" in LLM:

- Must define "correct output" (but generation has no correctness predicate)
- Must access "model's belief state" (but no belief states exist)
- Instrument (human judgment of falsity) doesn't separate from phenomenon (token generation process)

Result: $d \to 0$. We cannot separate measurement of "hallucination" from our own interpretation of outputs.

Scale separation (s):

At what scale does "hallucination" exist?

- Token level: Each token is most likely given context - no hallucination
- Sentence level: Grammatical, contextual - no hallucination
- Semantic level: Meaning is human interpretation - hallucination is observer-dependent
- Document level: Coherence is evaluator judgment - still observer-dependent

Result: $s \to 0$. No scale exists where "hallucination" is property of system rather than evaluator.

Diagnosis: This is not specification failure - it is structural measurement collapse. "Hallucination" is category error when applied to LLMs.

3.4 Layer 3: Access Asymmetry

The term creates cryptographic access structure:

Tier 1 (Training/architecture access):

- Researchers who understand that LLMs are likelihood maximizers, not truth-trackers
- Can see that "hallucination" is category error
- Have access to mechanistic explanation

Tier 2 (API/interface access):

- Engineers who work with model outputs
- See "hallucination" as "output that fails verification"
- Practical definition, but loses theoretical precision

Tier 3 (End-user access):

- General public receiving outputs
- Interpret "hallucination" through psychiatric metaphor
- Attribute mental states to models ("AI believes", "AI lies")
- Tier 3 users cannot access the meaning available at Tier 1

Power asymmetry:
The term "hallucination" creates interpretive lock-in:

- Tier 3 users develop model of AI as agent with faulty perception
- Tier 1 understanding ("likelihood maximization sometimes produces outputs humans judge false") is inaccessible
- Policy gets written using Tier 3 understanding ("prevent AI hallucinations")
- Technical impossibility follows (cannot prevent non-existent phenomenon)

3.5 Intervention: Γ-Correction

What would operative version look like?

1. **Specify measurement:** "LLM output judged factually incorrect by human evaluators"
2. **Specify frequency:** "X% of outputs on topic T fail fact-checking"
3. **Drop metaphor:** "Ungrounded generation" not "hallucination"
4. **Mechanistic description:** "High-likelihood generation that doesn't track truth"

Corrected claim:
"For task T, model M produces outputs that fail fact-checking with frequency F, due to likelihood maximization not being constrained by truth-tracking objectives."
This is:

- Operationally typeable ($\Phi = 1$)
- Measurable ($d > 0$, $s > 0$)
- Tier-transparent (mechanism available at all tiers)

3.6 Why This Matters

Current state:

- Billions spent on "reducing hallucinations"
- Regulatory frameworks designed around "hallucination risk"
- Public understanding locked into agentive model
- Technical community accepts imprecise metaphor for convenience

With typing:

- "Hallucination reduction" recognized as category error
- Focus shifts to: "Improving output-truth correlation for specified tasks"
- Public communication uses operative language
- Policy targets actual measurable properties

Stakes: This is not mere terminology. The metaphor shapes:

- Research directions (anthropomorphic goals)

- Safety frameworks (preventing non-existent phenomena)
- Public trust (expecting impossible guarantees)
- Legal liability (attributing mental states to statistical processes)

Conclusion: "LLMs hallucinate" is paradigmatic untypeable claim:

- Fails Φ-gate (no measurement instrument)
- Exhibits measurement collapse ($d \to 0$, $s \to 0$)
- Creates access asymmetry (Tier 3 locked out of Tier 1 understanding)
- Widely deployed in high-stakes contexts
- Γ-correctable to operative form

This demonstrates the framework's utility for addressing contemporary AI discourse.

3.7 Typed Evaluation for Language Models

The diagnostic framework presented in this paper suggests a broader implication for the evaluation of language models and other generative AI systems.

Current evaluation pipelines often assume that model outputs correspond to well-defined properties such as truthfulness, hallucination, or alignment. However, these predicates frequently lack operational typing conditions.

From the perspective of epistemic hygiene, many model evaluation benchmarks implicitly assume $\Phi(C) = 1$ without verifying that the claim is operationally typeable. This leads to systematic evaluation failures.

3.7.1 Evaluation as Typing Problem

Model evaluation can be reframed as a typing problem:

$$\begin{array}{c} \text{Model output } O \\ \downarrow \\ \text{Claim about property } P \\ \downarrow \\ \Phi\text{-gate} \\ \downarrow \\ \text{Operative predicate?} \end{array}$$

If $\Phi(P) \neq 1$, the evaluation protocol does not measure the property it claims to measure. Instead, it measures a proxy defined by evaluator interpretation.

Claim	**Typing problem**
"Model hallucination rate"	hallucination undefined
"Model truthfulness"	truth predicate undefined for generative distribution
"Model understanding"	no operational instrument

The framework therefore suggests a preliminary step for evaluation pipelines: **predicate typing before benchmark construction**.

3.7.2 Observation Algebra Expansion in AI Systems

Modern AI systems exhibit rapid expansion of observation algebras. A model evaluated under one observation regime may exhibit different properties under another.

Examples include:

- API-only access vs. parameter access
- Prompt evaluation vs. adversarial probing
- Benchmark testing vs. deployment interaction

Under expanded observation algebras, predicates previously considered valid may become untypeable. This phenomenon parallels the observation algebra relativity described in the Access Collapse framework (Isopahkala, 2026).

3.7.3 Toward Typed Benchmarks

If evaluation predicates are typed explicitly, benchmark construction becomes more robust.

Typed benchmark design would require:

1. **Explicit measurement instrument**

   Example: "Output judged factually incorrect by human evaluators using dataset D."

2. **Specified operational chain**

   prompt $\rightarrow$ model output $\rightarrow$ verification protocol $\rightarrow$ classification

3. **Declared access regime**

   e.g. API-only, white-box, or hybrid access

Benchmarks satisfying these conditions produce typeable evaluation claims. Benchmarks lacking them risk producing evaluation artifacts rather than measurable properties.

3.7.4 Implications for AI Safety

The typed evaluation perspective clarifies several persistent debates in AI safety:

Alignment claims: Statements such as "model is aligned with human values" often fail the Φ-gate due to undefined measurement conditions.

Hallucination mitigation: If hallucination is untypeable, research programs framed as "hallucination reduction" may target proxies rather than the underlying property.

Capability evaluation: Capability claims require explicit operational chains linking prompts, outputs, and task success.

The framework therefore suggests that many AI safety debates are not disagreements about truth but disagreements about typing conditions for evaluation predicates.

3.7.5 Research Direction

Future work could develop typed evaluation protocols for generative models. Such protocols would integrate:

- Operational typing (Φ-gate)
- Measurement regime specification
- Access structure transparency

This approach would shift evaluation from implicit semantic assumptions toward explicitly typed epistemic operations.

4 Limitations and Honest Boundaries

4.1 What This Framework Does Not Do

[Content continues...]

4.2 When NOT to Use This Framework

[Content continues...]

4.3 Self-Application

[Content continues...]

4.4 Validation: Preliminary Results and Path Forward

The framework requires empirical testing of intersubjective stability. We report preliminary results and specify the full validation protocol.

4.4.1 Preliminary Inter-Rater Study (Pilot)

Method:

- 3 independent analysts (1 philosopher, 1 computer scientist, 1 science communicator)
- 12 claims spanning 4 domains (AI safety, medical, political, scientific)
- Training: 2-hour workshop on framework + practice examples
- Task: Apply Φ-gate independently, classify as operative/non-operative/H_0
- Measure: Cohen's κ (agreement), qualitative comparison of reasoning

Claims tested (examples):

1. "Social media harms democracy" $\rightarrow$ Expected: $\Phi \neq 1$ (non-operative)
2. "Vaccine X reduces infection by 60% in trial population Y" $\rightarrow$ Expected: $\Phi = 1$
3. "Consciousness is quantum mechanical" $\rightarrow$ Expected: $\Phi \neq 1$ (category error)
4. "Algorithm discriminates against protected class" $\rightarrow$ Boundary case

Results:

Measure	Result	Interpretation
Overall agreement (κ)	0.68	Moderate/substantial agreement
Agreement on clear cases (1,2,3)	0.89	High convergence
Agreement on boundary case (4)	0.42	Low convergence (expected)
Convergence on Z-depth	± 1 level in 75%	Acceptable variance

Qualitative findings:

- High convergence when Δ-condition clearly satisfied/violated
- Divergence concentrated on "operative chain" assessment (condition 2)
- Analysts with domain expertise occasionally over-fitted (saw operative chains others didn't)

- Non-experts sometimes missed subtle measurement collapse

Interpretation:
The framework functions as shared diagnostic instrument for clear cases, but requires:

1. Protocol refinement for "operative chain" criterion
2. Calibration exercises for boundary cases
3. Acknowledgment that domain expertise shifts judgment (not necessarily error)

This is not falsification—it is protocol refinement need (Lakatosian progressive research program structure).

4.4.2 Full Validation Protocol

Needed for publication-grade validation:

Sample:

- Minimum 10 analysts (mix of domains: philosophy, science, policy, humanities)
- Mix of expertise levels (experts, educated non-experts)

Claims:

- 30 claims across 6 domains
- Include 10 calibration claims with ground truth
- 10 boundary cases where disagreement is expected
- 10 novel claims analysts haven't seen

Training:

- Standardized 4-hour workshop
- Practice set with discussion
- Written protocol document

Measures:

1. Cohen's κ for Φ-status (operative/non-operative/H_0)
2. Mean absolute deviation on Z-depth ($|Z_A - Z_B|$)
3. Jaccard similarity for access tier classifications
4. Qualitative: reasoning convergence (coded from written justifications)

Success criteria (pre-registered):

- $\kappa \geq 0.70$ overall (substantial agreement)
- $\kappa \geq 0.85$ on clear cases
- $|Z_A - Z_B| \leq 1$ in 80% of cases
- Jaccard ≥ 0.65 for access tiers

If criteria not met:

- Divergence patterns identify protocol gaps
- Refined protocol tested on new sample
- Iterative refinement (Lakatosian protective belt modification)

4.4.3 Current Status and Honest Limitations

Status: Preliminary pilot completed ($n = 3$). Full validation pending.

Until full validation:

- Framework is high-resolution methodological protocol
- Not yet validated inter-analyst instrument
- Not yet publication-ready as "objective diagnostic"

Honest assessment:

- Pilot suggests framework is more than personal tool
- Pilot suggests refinement needed, not fundamental rework
- Full validation is specifiable and tractable

We present framework now because:

1. Applied demonstrations have immediate value
2. Pedagogical tools are usable independent of perfect validation
3. Transparent limitations strengthen rather than weaken contribution
4. Framework can be useful while validation proceeds

This is intellectually honest positioning. We do not claim more than the evidence supports.

5 Conclusion

We have presented a three-layer diagnostic framework for epistemic hygiene:

Central contribution: Identification of untypeable claims as distinct structural pathology requiring distinct interventions. Most contemporary epistemic failures are not false claims but claims that have lost operational typing through:

- Reconstructability collapse (Φ-gate failure)
- Measurement condition violation ($d \to 0$ or $s \to 0$) (Isopahkala, 2026)
- Unanalyzed access asymmetry (tier structure invisible)

The framework integrates:

- Layer 1 (NAEI): Separates operative from non-operative claims
- Layer 2 (Phenomenosophy): Identifies measurement collapse conditions
- Layer 3 (Cryptosophy): Reveals access asymmetries and structural power

Demonstrated through four applied cases:

- AI safety (type error boundaries in query-based access)
- Scientific communication (Project A/B confusion in expertise)
- Public deliberation (epistemic symmetry failures)
- Medicalization (diagnostic categories as cryptographic gates)

Positioned relative to existing work:

- Extends Collins & Evans on expertise with formal access tiers
- Operationalizes Fricker's epistemic injustice with structural diagnostics
- Formalizes Conrad's medicalization as cryptographic mechanism
- Type-theoretically grounds Tramèr et al.'s extraction attacks

Validated (preliminarily):

- Pilot study ($n = 3$) shows moderate inter-rater agreement ($\kappa = 0.68$)
- Suggests framework can function as shared instrument with protocol refinement
- Full validation path specified and tractable

Pedagogical tools provided:

- Diagnostic checklists for practitioners
- Recognition exercises for common errors
- Intervention strategies for each failure mode

The framework enables:

- **Precision without dogma:** Claims can be non-operative without being false
- **Boundaries without ideology:** Measurement limits are structural, not normative
- **Power analysis without paranoia:** Access asymmetries can be functional if visible

Core principle: Not everything that can be said should be said as knowledge claim. Not everything that can be measured should be measured in every regime. Not everything that fails these tests is wrong—it may simply be untypeable in its current operational context.

The appropriate response to untypeable claims is not refutation but **retyping**—Γ-correction that restores operative status or honest reclassification as H_0 (exploratory, not yet testable).

This framework is offered as diagnostic instrument, not theory of truth. Its ultimate validation depends on whether independent analysts converge on its application across contexts—a test that preliminary results suggest is achievable with protocol refinement.

Appendix A: Preliminary Validation Data

Table 1: Inter-Rater Agreement by Claim Type

Claim	**A1**	**A2**	**A3**	κ
"Social media harms democracy"	$\neq 1$	$\neq 1$	$\neq 1$	1.00
"Vaccine reduces infection 60%"	1	1	1	1.00
"Consciousness is quantum"	$\neq 1$	$\neq 1$	H_0	0.67
"Algorithm discriminates"	$\neq 1$	1*	$\neq 1$	0.33

*Analyst 2 (computer scientist) specified implicit measurement: "disparate impact ≥ 1.25 ratio" and judged operative with this specification. Illustrates domain expertise effect.

Key finding: Z-depth variance acceptable (≤ 1 level) in 9/12 cases.

Table 2: Z-Depth Convergence

Claim	**A1**	**A2**	**A3**	**StdDev**
"AI harms thinking" (uncorrected)	2	3	2	0.58
"Vaccine 60% effective"	1	1	1	0.00
"Consciousness quantum"	3	4	3	0.58

References

Collins, H. and Evans, R. (2007). *Rethinking Expertise.* University of Chicago Press, Chicago.

Conrad, P. (2007). *The Medicalization of Society: On the Transformation of Human Conditions into Treatable Disorders.* Johns Hopkins University Press, Baltimore.

Fricker, M. (2007). *Epistemic Injustice: Power and the Ethics of Knowing.* Oxford University Press, Oxford.

Isopahkala, A. (2026). Access collapse: A typed structural boundary for operational access. Our foundational theoretical treatment establishing the Access Collapse Theorem, Observation Algebra Relativity, and formal conditions for measurement type errors. Provides the formal architecture for the d (operational separateness) and s (scale separation) parameters used throughout this framework.

Jagielski, M., Carlini, N., Berthelot, D., Kurakin, A., and Papernot, N. (2020). High accuracy and high fidelity extraction of neural networks. In *Proceedings of the 29th USENIX Security Symposium*, pages 1345–1362.

Tramèr, F., Zhang, F., Juels, A., Reiter, M. K., and Ristenpart, T. (2016). Stealing machine learning models via prediction apis. In *Proceedings of the 25th USENIX Security Symposium*, pages 601–618.